Automated Deployment of Oracle 19c for ONTAP on NFS

NetApp Solutions

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Solution Overview

Automated Deployment of Oracle19c for ONTAP on NFS

Organizations are automating their environments to gain efficiencies, accelerate deployments, and reduce manual effort. Configuration management tools like Ansible are being used to streamline enterprise database operations. In this solution, we demonstrate how you can use Ansible to automate the provisioning and configuration of Oracle 19c with NetApp ONTAP. By enabling storage administrators, systems administrators, and DBAs to consistently and rapidly deploy new storage, configure database servers, and install Oracle 19c software, you achieve the following benefits:

- Eliminate design complexities and human errors, and implement a repeatable consistent deployment and best practices
- Decrease time for provisioning of storage, configuration of DB hosts, and Oracle installation
- Increase database administrators, systems and storage administrators productivity
- Enable scaling of storage and databases with ease

NetApp provides customers with validated Ansible modules and roles to accelerate deployment, configuration, and lifecycle management of your Oracle database environment. This solution provides instruction and Ansible playbook code, to help you:

- Create and configure ONTAP NFS storage for Oracle Database
- Install Oracle 19c on RedHat Enterprise Linux 7/8 or Oracle Linux 7/8
- Configure Oracle 19c on ONTAP NFS storage

For more details or to begin, please see the overview videos below.

AWX/Tower Deployments

- Part 1: Getting Started, Requirements, Automation Details and Initial AWX/Tower Configuration

▶ https://docs.netapp.com/us-en/netapp-solutions/media/oracle_deployment_auto_v1.mp4 *(video)*

- Part 2: Variables and Running the Playbook

▶ https://docs.netapp.com/us-en/netapp-solutions/media/oracle_deployment_auto_v2.mp4 *(video)*

CLI Deployment

- Part 1: Getting Started, Requirements, Automation Details and Ansible Control Host Setup

▶ https://docs.netapp.com/us-en/netapp-solutions/media/oracle_deployment_auto_v4.mp4 *(video)*

- Part 2: Variables and Running the Playbook

▶ https://docs.netapp.com/us-en/netapp-solutions/media/oracle3.mp4 *(video)*

Getting started

This solution has been designed to be run in an AWX/Tower environment or by CLI on an Ansible control host.

AWX/Tower

For AWX/Tower environments, you are guided through creating an inventory of your ONTAP cluster management and Oracle server (IPs and hostnames), creating credentials, configuring a project that pulls the Ansible code from NetApp Automation Github, and the Job Template that launches the automation.

1. Fill out the variables specific to your environment, and copy and paste them into the Extra Vars fields in your job template.
2. After the extra vars have been added to your job template, you can launch the automation.
3. The job template is run in three phases by specifying tags for ontap_config, linux_config, and oracle_config.

CLI via the Ansible control host

1. To configure the Linux host so that is can be used as an Ansible control host
   click here for RHEL 7/8 or CentOS 7/8, or
   here for Ubuntu/Debian
2. After the Ansible control host is configured, you can git clone the Ansible Automation repository.
3. Edit the hosts file with the IPs and/or hostnames of your ONTAP cluster management and Oracle server's management IPs.
4. Fill out the variables specific to your environment, and copy and paste them into the `vars.yml` file.
5. Each Oracle host has a variable file identified by its hostname that contains host-specific variables.
6. After all variable files have been completed, you can run the playbook in three phases by specifying tags for `ontap_config`, `linux_config`, and `oracle_config`.

Requirements

Environment	Requirements
Ansible environment	AWX/Tower or Linux host to be the Ansible control host
	Ansible v.2.10 and higher
	Python 3
	Python libraries - netapp-lib - xmltodict - jmespath
ONTAP	ONTAP version 9.3 - 9.7
	Two data aggregates
	NFS vlan and ifgrp created

Environment	Requirements
Oracle server(s)	RHEL 7/8
	Oracle Linux 7/8
	Network interfaces for NFS, public, and optional mgmt
	Oracle installation files on Oracle servers

Automation Details

This automated deployment is designed with a single Ansible playbook that consists of three separate roles. The roles are for ONTAP, Linux, and Oracle configurations.
The following table describes which tasks are being automated.

Role	Tasks
ontap_config	Pre-check of the ONTAP environment
	Creation of NFS based SVM for Oracle
	Creation of export policy
	Creation of volumes for Oracle
	Creation of NFS LIFs
linux_config	Create mount points and mount NFS volumes
	Verify NFS mounts
	OS specific configuration
	Create Oracle directories
	Configure hugepages
	Disable SELinux and firewall daemon
	Enable and start chronyd service
	increase file descriptor hard limit
	Create pam.d session file
oracle_config	Oracle software installation
	Create Oracle listener
	Create Oracle databases
	Oracle environment configuration
	Save PDB state
	Enable instance archive mode
	Enable DNFS client
	Enable database auto startup and shutdown between OS reboots

Default parameters

To simplify automation, we have preset many required Oracle deployment parameters with default values. It is generally not necessary to change the default parameters for most deployments. A more advanced user can make changes to the default parameters with caution. The default parameters are located in each role folder under defaults directory.

Deployment instructions

Before starting, download the following Oracle installation and patch files and place them in the `/tmp/archive` directory with read, write, and execute access for all users on each DB server to be deployed. The automation tasks look for the named installation files in that particular directory for Oracle installation and configuration.

```
LINUX.X64_193000_db_home.zip -- 19.3 base installer
p31281355_190000_Linux-x86-64.zip -- 19.8 RU patch
p6880880_190000_Linux-x86-64.zip -- opatch version 12.2.0.1.23
```

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After you are ready, click here for detailed AWX/Tower deployment procedures or here for CLI deployment.

Step-by-step deployment procedure

AWX/Tower deployment Oracle 19c Database

1. Create the inventory, group, hosts, and credentials for your environment

This section describes the setup of inventory, groups, hosts, and access credentials in AWX/Ansible Tower that prepare the environment for consuming NetApp automated solutions.

1. Configure the inventory.
    a. Navigate to Resources → Inventories → Add, and click Add Inventory.
    b. Provide the name and organization details, and click Save.
    c. On the Inventories page, click the inventory created.
    d. If there are any inventory variables, paste them in the variables field.
    e. Navigate to the Groups sub-menu and click Add.
    f. Provide the name of the group for ONTAP, paste the group variables (if any) and click Save.
    g. Repeat the process for another group for Oracle.
    h. Select the ONTAP group created, go to the Hosts sub-menu and click Add New Host.

i. Provide the IP address of the ONTAP cluster management IP, paste the host variables (if any), and click Save.

j. This process must be repeated for the Oracle group and Oracle host(s) management IP/hostname.

2. Create credential types. For solutions involving ONTAP, you must configure the credential type to match username and password entries.

   a. Navigate to Administration → Credential Types, and click Add.

   b. Provide the name and description.

   c. Paste the following content in Input Configuration:

```
fields:
  - id: username
    type: string
    label: Username
  - id: password
    type: string
    label: Password
    secret: true
  - id: vsadmin_password
    type: string
    label: vsadmin_password
    secret: true
```

   d. Paste the following content into Injector Configuration:

```
extra_vars:
  password: '{{ password }}'
  username: '{{ username }}'
  vsadmin_password: '{{ vsadmin_password }}'
```

3. Configure the credentials.

   a. Navigate to Resources → Credentials, and click Add.

   b. Enter the name and organization details for ONTAP.

   c. Select the custom Credential Type you created for ONTAP.

   d. Under Type Details, enter the username, password, and vsadmin_password.

   e. Click Back to Credential and click Add.

   f. Enter the name and organization details for Oracle.

   g. Select the Machine credential type.

   h. Under Type Details, enter the Username and Password for the Oracle hosts.

   i. Select the correct Privilege Escalation Method, and enter the username and password.

2. Create a project

1. Go to Resources → Projects, and click Add.
   a. Enter the name and organization details.
   b. Select Git in the Source Control Credential Type field.
   c. enter https://github.com/NetApp-Automation/na_oracle19c_deploy.git as the source control URL.
   d. Click Save.
   e. The project might need to sync occasionally when the source code changes.

3. Configure Oracle host_vars

The variables defined in this section are applied to each individual Oracle server and database.

1. Input your environment-specific parameters in the following embedded Oracle hosts variables or host_vars form.

   The items in blue must be changed to match your environment.

Unresolved directive in ent-apps-db/awx_automation.adoc - include::ent-apps-db/host_vars.adoc[]

a. Fill in all variables in the blue fields.
b. After completing variables input, click the Copy button on the form to copy all variables to be transferred to AWX or Tower.
c. Navigate back to AWX or Tower and go to Resources → Hosts, and select and open the Oracle server configuration page.
d. Under the Details tab, click edit and paste the copied variables from step 1 to the Variables field under the YAML tab.
e. Click Save.
f. Repeat this process for any additional Oracle servers in the system.

4. Configure global variables

Variables defined in this section apply to all Oracle hosts, databases, and the ONTAP cluster.

1. Input your environment-specific parameters in following embedded global variables or vars form.

   The items in blue must be changed to match your environment.

   Unresolved directive in ent-apps-db/awx_automation.adoc - include::ent-apps-db/vars.adoc[]

2. Fill in all variables in blue fields.
3. After completing variables input, click the Copy button on the form to copy all variables to be transferred to AWX or Tower into the following job template.

5. Configure and launch the job template.

1. Create the job template.
   a. Navigate to Resources → Templates → Add and click Add Job Template.

b. Enter the name and description
c. Select the Job type; Run configures the system based on a playbook, and Check performs a dry run of a playbook without actually configuring the system.
d. Select the corresponding inventory, project, playbook, and credentials for the playbook.
e. Select the all_playbook.yml as the default playbook to be executed.
f. Paste global variables copied from step 4 into the Template Variables field under the YAML tab.
g. Check the box Prompt on Launch in the Job Tags field.
h. Click Save.

2. Launch the job template.
   a. Navigate to Resources → Templates.
   b. Click the desired template and then click Launch.
   c. When prompted on launch for Job Tags, type in requirements_config. You might need to click the Create Job Tag line below requirements_config to enter the job tag.

> requirements_config ensures that you have the correct libraries to run the other roles.

   d. Click Next and then Launch to start the job.
   e. Click View → Jobs to monitor the job output and progress.
   f. When prompted on launch for Job Tags, type in ontap_config. You might need to click the Create "Job Tag" line right below ontap_config to enter the job tag.
   g. Click Next and then Launch to start the job.
   h. Click View → Jobs to monitor the job output and progress
   i. After the ontap_config role has completed, run the process again for linux_config.
   j. Navigate to Resources → Templates.
   k. Select the desired template and then click Launch.
   l. When prompted on launch for the Job Tags type in linux_config, you might need to select the Create "job tag" line right below linux_config to enter the job tag.
   m. Click Next and then Launch to start the job.
   n. Select View → Jobs to monitor the job output and progress.
   o. After the linux_config role has completed, run the process again for oracle_config.
   p. Go to Resources → Templates.
   q. Select the desired template and then click Launch.
   r. When prompted on launch for Job Tags, type oracle_config. You might need to select the Create "Job Tag" line right below oracle_config to enter the job tag.
   s. Click Next and then Launch to start the job.
   t. Select View → Jobs to monitor the job output and progress.

6. Deploy additional database on same Oracle host

The Oracle portion of the playbook creates a single Oracle container database on an Oracle server per execution. To create additional container databases on the same server, complete the following steps.

1. Revise host_vars variables.
    a. Go back to step 2 - Configure Oracle host_vars.
    b. Change the Oracle SID to a different naming string.
    c. Change the listener port to different number.
    d. Change the EM Express port to a different number if you are installing EM Express.
    e. Copy and paste the revised host variables to the Oracle Host Variables field in the Host Configuration Detail tab.
2. Launch the deployment job template with only the oracle_config tag.

Unresolved directive in ent-apps-db/awx_automation.adoc - include::ent-apps-db/validation.adoc[]

Step-by-step deployment procedure

CLI deployment Oracle 19c Database

This section covers the steps required to prepare and deploy Oracle19c Database with the CLI. Make sure that you have reviewed the Getting Started and Requirements section and prepared your environment accordingly.

Download Oracle19c repo

```
1. From your ansible controller, run the following command:
```

```
git clone https://github.com/NetApp-Automation/na_oracle19c_deploy.git
```

```
2. After downloading the repository, change directories to
na_oracle19c_deploy <cd na_oracle19c_deploy>.
```

Edit the hosts file

Complete the following before deployment:

1. Edit your hosts file na_oracle19c_deploy directory.
2. Under [ontap], change the IP address to your cluster management IP.
3. Under the [oracle] group, add the oracle hosts names. The host name must be resolved to its IP address either through DNS or the hosts file, or it must be specified in the host.
4. After you have completed these steps, save any changes.

The following example depicts a host file:

```
#ONTAP Host<div>
[ontap]
<div>
<span <div contenteditable="false" style="color:#7EAF97
; font-weight:bold; font-style:italic; text-
decoration:;"/>10.61.184.183<i></i></span>
</div>
#Oracle hosts<div>
<div>
[oracle]<div>
<span <div contenteditable="false" style="color:#7EAF97
; font-weight:bold; font-style:italic; text-
decoration:;"/>rtpora01<i></i></span>
<div>
<span <div contenteditable="false" style="color:#7EAF97
; font-weight:bold; font-style:italic; text-
decoration:;"/>rtpora02<i></i></span>
</div>
```

This example executes the playbook and deploys oracle 19c on two oracle DB servers concurrently. You can also test with just one DB server. In that case, you only need to configure one host variable file.

> The playbook executes the same way regardless of how many Oracle hosts and databases you deploy.

Edit the host_name.yml file under host_vars

Each Oracle host has its host variable file identified by its host name that contains host-specific variables. You can specify any name for your host. Edit and copy the `host_vars` from the Host VARS Config section and paste it into your desired `host_name.yml` file.

> The items in blue must be changed to match your environment.

Unresolved directive in ent-apps-db/cli_automation.adoc - include::ent-apps-db/host_vars.adoc[]

Edit the vars.yml file

The `vars.yml` file consolidates all environment-specific variables (ONTAP, Linux, or Oracle) for Oracle deployment.

- Edit and copy the variables from the VARS section and paste these variables into your `vars.yml` file.

Unresolved directive in ent-apps-db/cli_automation.adoc - include::ent-apps-db/vars.adoc[]

Run the playbook

After completing the required environment prerequisites and copying the variables into `vars.yml` and `your_host.yml`, you are now ready to deploy the playbooks.

<username> must be changed to match your environment.

```
1. Run the ONTAP playbook by passing the correct tags and ONTAP cluster
username. Fill the password for ONTAP cluster, and vsadmin when prompted.
```

```
ansible-playbook -i hosts all_playbook.yml -u username -k -K -t
ontap_config -e @vars/vars.yml
```

```
2. Run the Linux playbook to execute Linux portion of deployment. Input
for admin ssh password as well as sudo password.
```

```
ansible-playbook -i hosts all_playbook.yml -u username -k -K -t
linux_config -e @vars/vars.yml
```

```
3. Run the Oracle playbook to execute Oracle portion of deployment. Input
for admin ssh password as well as sudo password.
```

```
ansible-playbook -i hosts all_playbook.yml -u username -k -K -t
oracle_config -e @vars/vars.yml
```

Deploy Additional Database on Same Oracle Host

The Oracle portion of the playbook creates a single Oracle container database on an Oracle server per execution. To create additional container database on the same server, complete the following steps:

1. Revise the host_vars variables.
   a. Go back to step 3 - Edit the `host_name.yml` file under `host_vars`.
   b. Change the Oracle SID to a different naming string.
   c. Change the listener port to different number.
   d. Change the EM Express port to a different number if you have installed EM Express.
   e. Copy and paste the revised host variables to the Oracle host variable file under `host_vars`.
2. Execute the playbook with the `oracle_config` tag as shown above in Run the playbook.

Unresolved directive in ent-apps-db/cli_automation.adoc - include::ent-apps-db/validation.adoc[]

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